EAT40005 Engineering Technology Projects A

Individual Work Log

PROJECT NAME:	Whole-Body Model Predictive Control for Industrial Humanoids
STUDENT NAME:	**Rumeth Gunasinghe**
STUDENT ID:	**104498227**

Cumulative Summary of week 8

Week 3	Major work done this week	Total hours spent this week	Cumulative Total
1	**Migrated project focus from PyBullet humanoid to MuJoCo simulation environment**	**4**	10
2	**Researched MuJoCo architecture, XML models, actuators, sensors and control loop structure**	5	11
3	**Designed simplified humanoid robot structure with reduced degrees of freedom**	3	7
4	**Planned locomotion system for robot to move toward target object**	**3**	**9**
5	**Planned object interaction sequence (reach, grasp, lift)**	**2**	**9**
6	**Team meeting and Sprint 2 planning discussion**	**1.5**	**3**
7	**Updated modular project architecture for MuJoCo implementation**	**3**	**5**

Week 1

WEEK #	1		
Dates covered:	**02/03/2026 – 08/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Researching on potential projects	Completed	**10**	**Asking for personal project proposal**
Communicating with the convenor	completed	**4**	**Choosing a given project from the list**
TOTAL WEEKLY TIME SPENT		**14**	

Summary/weekly reflection for Week 1:

- Researched on available projects to find the suitable one
- Communicating with the convenor for support

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
More research into available projects	15/03/2026
osing a suitable project	15/03/2026

Week 2

WEEK #	2		
Dates covered:	**09/03/2026 – 15/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Researching on projects	Completed	**8**	**Chosing the humanoid robot control project**
Researching on the selected project	Completed	**12**	**Drafting a plan**
TOTAL WEEKLY TIME SPENT		**20**	

Summary/weekly reflection for Week 2:

- Researching on available projects
- Researching on the selected project

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Finding a Teammate	22/03/2026
Finishing up the project and team plan assignment	22/03/2026

Week 3

WEEK #	3		
Dates covered:	**16/-3/2026 – 22/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Project allocation and understanding project requirements	Completed	**2**	**Task devision among members**
Research on simulation tools (PyBullet / MuJoCo)	Completed	**5**	**Installed necessary libraries**
Finshing assignment 1	Completed	8	Submitted the assignment
TOTAL WEEKLY TIME SPENT		**15**	

Summary/weekly reflection for Week 3:

- Project was assigned and understood
- Team roles and responsibilities were defined
- Initial research on humanoid robotics and control systems conducted
- Completed major sections of the project proposal/report
- Set up tools for development (GitHub, communication platforms)

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Working on Assignment 2 (research report)	29/03/2026
Researching and start working on the project implementation	On going

Week 4

WEEK 4	4		
Dates covered:	**23/03/2026 – 29/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Completed research document	Completed	5	Research doc submission
Continued research on Model Predictive Control (MPC)	On going	**4**	drafts
Research on inverted pendulum model	On going	4	drafts
Planning system architecture	Completed	**3**	Finalised System Architecture
TOTAL WEEKLY TIME SPENT		**16**	

Summary/weekly reflection for Week #:

- Conducted in-depth research on MPC and control systems
- Identified inverted pendulum as the correct modelling approach
- Defined high-level system architecture
- Started research report writing
- Set up Jira sprint structure

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Begin implementation of inverted pendulum simulation	05/04/2026
Testing simulation Behavior	05/04/2026

Week 5

WEEK #	5		
Dates covered:	**30/03/2026 - 12/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Studied PID control fundamentals	Completed	3	Understood feedback control and tuning parameters
Started inverted pendulum simulation implementation	Completed	4	Implemented basic system model in Python
Tested simulation behaviour	Completed	2	Verified system response under simple conditions
Further research on MPC implementation methods	Completed	2	Focused on optimisation-based control
TOTAL WEEKLY TIME SPENT		11	

Summary/weekly reflection for Week #:

- *Studied PID control as baseline method*
- *Started implementation of inverted pendulum simulation*
- *Conducted initial testing of system behaviour*
- Difference between reactive (PID) and predictive (MPC) control
- Importance of feedback loops in maintaining system stability
- Simulation helps validate concepts before complex implementation

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Working on Sprint report 1	19/04/2026
Implementing and testing on PID and MPC	19/04/2026

Week 6

WEEK #	6		
Dates covered:	**12 /04/2026 – 19/04/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Implemented PID controller with full cart-pole dynamics	Completed	5	Integrated PID with realistic physics (cart + pole interaction)
Implemented MPC controller with prediction and cost function	Completed	4	Developed simulate_step() and cost-based optimisation
Fixed instability issues in simulation	Completed	2	Resolved issues like drifting, unrealistic motion, and imbalance
Implemented data logging and performance graphs	Completed	2	Generated theta, position, and force graphs
TOTAL WEEKLY TIME SPENT		11	

Summary/weekly reflection for Week #:

- Successfully implemented both PID and MPC controllers using realistic cart-pole dynamics
- Understood how system states (position, velocity, angle) are interconnected
- Learned that controller performance depends heavily on tuning (gains for PID, cost function for MPC)
- Observed differences between PID (reactive) and MPC (predictive) in handling disturbances
- Identified issues such as drift and continuous movement due to lack of position constraints
- Verified that the cart moves in response to pole angle, confirming correct physical coupling
- Introduced disturbance testing to simulate real-world conditions

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Plan and organise Sprint 2	26/04/2026
Implementing and testing MPC humanoid simulations	26/04/2026

Week 7

WEEK #	7		
Dates covered:	**19 /04/2026 – 26/05/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Implemented PID controller with full cart-pole dynamics	Completed	5	Developed base simulation using PyBullet
Implemented MPC controller with prediction and cost function	Completed	4	Resolved Not connected to physics server error
Fixed instability issues in simulation	Completed	2	Verified joint control and motion responses
Implemented data logging and performance graphs	Completed	2	Improved modular structure for scalability
TOTAL WEEKLY TIME SPENT		13	

Summary/weekly reflection for Week #:

- Focused on implementing the core simulation environment
- Successfully resolved critical PyBullet connection issues
- Tested initial humanoid robot movement and control
- Improved overall code structure and modularity
- Gained better understanding of physics-based simulation challenges

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Integrating AI control logic (basic decision-making)	03/05/2026
Enhancing robot movement with smoother transitions	03/05/2026
Begin implementing environment interaction (obstacles/tasks)	03/05/2026

Week 8

WEEK #	7		
Dates covered:	**26 /04/2026 – 03/05/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Migrated project focus from PyBullet humanoid to MuJoCo simulation environment	Completed	5	Reviewed updated client requirements and selected MuJoCo for locomotion/object interaction tasks
Researched MuJoCo architecture, XML models, actuators, sensors and control loop structure	Completed	4	Investigated MuJoCo workflow for humanoid robotics and simplified model control
Designed simplified humanoid robot structure with reduced degrees of freedom	Completed	2	Planned leg joints, torso, arm joints and gripper for efficient implementation
Planned locomotion system for robot to move toward target object	Completed	2	Developed walking logic using scripted gait / control state transitions
Planned object interaction sequence (reach, grasp, lift)	Completed	3	Defined task stages for navigation and manipulation behaviour
Updated modular project architecture for MuJoCo implementation	Completed	2	Refined folders for models, controllers, tasks and utilities
TOTAL WEEKLY TIME SPENT		13	

Summary/weekly reflection for Week #:

- Project direction transitioned from balancing-only control to full humanoid task execution in MuJoCo.
- Gained strong understanding of MuJoCo simulation workflow, XML robot modelling and actuator control systems.
- Planned a simplified humanoid robot to reduce complexity while still demonstrating locomotion and manipulation.
- Designed behaviour flow for robot to walk to a target, stop accurately, reach an object and lift it.
- Improved understanding of how previous PID/MPC knowledge transfers into movement and interaction tasks.

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Integrating AI control logic (basic decision-making)	03/05/2026
Enhancing robot movement with smoother transitions	03/05/2026
Begin implementing environment interaction (obstacles/tasks)	03/05/2026